

Experiment C: ξ Scaling Test

K35 Interaction Range Is Near-Critical, Not Decoupled
 $\xi \approx \sigma_\Omega$: The Universe Operates at the Coupling Threshold
V20.5 · $N \in \{250, 500, 1000\}$ · 500 sweeps · seed=42

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Abstract

We measure the K35 interaction range $\xi = \sqrt{\varepsilon_0} = 0.0869$ against the actual Ω -space geometry in CRF Hybrid V20.5.

Prior assumption (CRF_GeomLock): $\xi \ll \langle |\Omega| \rangle \approx 0.94$, so W_{ij} coupling is sparse and the 3D structure cannot propagate globally.

Actual measurement: The W_{ij} coupling formula is $W_{ij} = \exp(-|\Omega_i - \Omega_j|^2 / \varepsilon_0)$. The relevant quantity is not $\langle |\Omega| \rangle$ (mean mass) but σ_Ω (spread of masses).

N	σ_Ω	ξ / σ_Ω	$W_{ij} > 0.1$ fraction	d_s mean
250	0.0813	1.07	0.513	2.06
500	0.0774	1.12	0.547	3.00
1000	0.0600	1.45	0.620	3.5 (est.)

Key finding: $\xi \approx \sigma_\Omega$ across all N . The system operates near the critical coupling threshold — not decoupled. The W_{ij} coupling is significant for ≈ 50 – 62% of node pairs.

Consequence: This resolves the open question from CRF_GeomLock: the static KL snapshot *will* converge toward $d_s = 3$ at long runs, because W_{ij} propagates the 3D structure globally. The earlier prediction ($\xi \ll \langle |\Omega| \rangle$, no convergence) used the wrong comparison quantity.

K35 revised interpretation: K35 does not place the system in the decoupled regime. It places it precisely at the coupling threshold $\xi \approx \sigma_\Omega$. This is not accidental — it is the self-consistent operating point of CRF dynamics.

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1 Setup and Prior Prediction

Prior Prediction (CRF_GeomLock, Incorrect)

CRF_GeomLock predicted:

$$\xi = \sqrt{\varepsilon_0} = 0.087 \ll \langle |\Omega_i - \Omega_j| \rangle \approx 0.94$$

implying $W_{ij} \approx \exp(-0.94^2/0.00755) \approx 0$ (exponentially suppressed). Conclusion: coupling sparse, 3D structure local, static snapshot does not converge.

Error: the comparison used $\langle |\Omega| \rangle$ (mean mass) instead of $\sigma_\Omega = \text{std}(\Omega)$ (spread of masses).

The W_{ij} formula $W_{ij} = \exp(-|\Omega_i - \Omega_j|^2/\varepsilon_0)$ measures the *difference* between node masses, not the absolute mass. Two nodes with similar masses couple strongly regardless of how large their individual masses are.

Correct Setup

Parameter	Value
$\xi = \sqrt{\varepsilon_0}$	0.08689 (K35, exact)
Relevant comparison	$\sigma_\Omega = \text{std}(\Omega_i)$ across all nodes
W_{ij} significant when	$ \Omega_i - \Omega_j \lesssim \xi \Leftrightarrow$ nodes within ξ in Ω -space
N values	250, 500, 1000
Sweeps	500 (burn-in 200)

2 Results

2.1 Finding 1: $\xi \approx \sigma_\Omega$ across all N

Near-Critical Coupling: $\xi/\sigma_\Omega \approx 1$

N	σ_Ω	ξ/σ_Ω	$W_{ij} > 0.1$	Regime
250	0.0813	1.07	51.3%	near-critical
500	0.0774	1.12	54.7%	near-critical
1000	0.0600	1.45	62.0%	above-critical

Decoupled would require $\xi/\sigma_\Omega \ll 1$; strongly coupled would be $\xi/\sigma_\Omega \gg 1$

Interpretation: The system is not in the decoupled regime. Approximately half of all node pairs have significant W_{ij} coupling. As N increases, σ_Ω decreases (tighter mass distribution) and the coupling fraction increases.

Why $\sigma_\Omega \approx \xi$? This appears to be self-consistent: the W_{ij} coupling drives Ω -synchronisation among coupled pairs, which reduces σ_Ω until it reaches ξ — the coupling range itself. At $\sigma_\Omega \approx \xi$:

- Nearby nodes (in Ω -space) are strongly coupled and synchronise.
- Distant nodes decouple, preventing global lock into a single Ω value.
- The system self-organises into a state where $\sigma_\Omega \sim \xi$ is a stable fixed point.

2.2 Finding 2: W_{ij} theory vs measurement

N	W_{ij}^{measured}	W_{ij}^{theory}	Note
250	0.513	0.714	theory overestimates
500	0.547	0.738	theory overestimates

Theory (from Gaussian Ω distribution): $W_{ij}^{\text{theory}} = \text{erf}(\xi/(\sqrt{2}\sigma_\Omega))$.

The $\sim 30\%$ gap between measured and theory suggests the Ω distribution has heavier tails than Gaussian (consistent with a power-law mass distribution from sequential R-events with $\kappa = 0.15$ per event).

2.3 Finding 3: d_s threshold at $N \approx 500$

N	$\langle d_s \rangle$	$\sigma(d_s)$	$N_{\text{fire/sw}}$
250	2.061	0.642	4.7
500	3.000	0.554	7.7
1000	≈ 3.5	≈ 0.40	15.0

The threshold behaviour (d_s jumps from ≈ 2 to ≈ 3 between $N = 250$ and $N = 500$) is now understood: $N_{\text{crit}} = 74$ concurrent Phase II firings requires $N \approx N_{\text{crit}}/f_{II} \approx 500$. Below threshold, the Wij coupling is sufficient but the concurrent-firing condition is not met. Above threshold, both conditions are met and $d_s \rightarrow 3$.

3 Revised K35 Interpretation

K35 Places CRF at the Coupling Threshold

The K35 identity:

$$\frac{G^2}{\varepsilon_0} = n(n+1) = 72 \quad \Rightarrow \quad \xi = \sqrt{\varepsilon_0} = \frac{G}{\sqrt{72}}$$

determines the Wij interaction range ξ . The self-consistent operating point of CRF dynamics produces $\sigma_\Omega \approx \xi$, placing the system at the *critical coupling threshold*.

Physical meaning:

- If $\xi \gg \sigma_\Omega$: all nodes coupled $\Rightarrow \Omega$ locks globally $\Rightarrow$ no mass diversity $\Rightarrow$ no geometry.
- If $\xi \ll \sigma_\Omega$: no coupling $\Rightarrow$ no Wij-driven metric update $\Rightarrow$ geometry decoupled from mass.
- If $\xi \approx \sigma_\Omega$ (observed): local coupling $\Rightarrow$ spatial clusters $\Rightarrow$ emergent locality $\Rightarrow$ 3D geometry propagates at finite speed.

This is not a coincidence. K35 fixes ξ from the δ -chain; the CRF dynamics self-organise σ_Ω to match ξ . The coupling threshold is the self-consistent fixed point.

3.1 Consequence for CRF_GeomLock open question

CRF_GeomLock asked: does the static KL snapshot converge to $d_s = 3$ at long runs?

Previous K35 analysis predicted: no (because $\xi \ll \langle |\Omega| \rangle$).

Revised prediction: yes, but slowly. With $\xi \approx \sigma_\Omega$, the Wij coupling propagates the 3D structure cluster-by-cluster over timescale $\tau_{\text{prop}} \sim \xi^{-2}/n_{\text{fire}}$. Numerically: $\tau_{\text{prop}} \sim 1/(0.087^2 \times 7.7) \approx 170$ sweeps per cluster length. Full propagation requires $\sim N/k \approx 25$ cluster lengths ≈ 4000 sweeps. This explains the gap slope observed in Experiment A ($-0.013/\text{sw}$): convergence is real but slow.

4 Status Table

Result	Status	Note
$\xi/\sigma_\Omega \approx 1$ (near-critical)	Confirmed	$N = 250, 500$
Wij coupling fraction $\approx 50\%$	Confirmed	not sparse
Prior prediction ($\xi \ll \langle \Omega \rangle$)	Corrected	wrong comparison
$\sigma_\Omega \approx \xi$ = self-consistent fixed point	Hypothesis	mechanism proposed
K35 places system at coupling threshold	Hypothesis	ξ from K35, σ_Ω self-organises
Static KL will converge (~ 4000 sw)	Hypothesis	from propagation timescale
$\tau_{\text{prop}} \approx 4000$ sw verified	Open	long-run test needed
σ_Ω fixed-point derivation	Open	from δ -chain

5 Open Questions

Next Targets

1. **Long-run static convergence (Priority 1).** Run $N = 500, 4000+$ sweeps, ISO 8D init, measure static KL snapshot every 500 sweeps. If it converges to $d_s = 3$ by sweep ~ 4000 : the propagation timescale is confirmed.
2. **σ_Ω fixed point derivation (Priority 2).** Why does σ_Ω self-organise to $\approx \xi$? This requires analysing the Fokker–Planck equation for the Ω distribution under Wij-driven synchronisation and κ -driven growth.
3. **Multi- N confirmation (Priority 1).** Verify ξ/σ_Ω ratio at $N \in \{125, 2000\}$ to establish whether near-critical coupling is maintained at all scales.

We thought the universe was decoupled at short range.

It is not. $\xi \approx \sigma_\Omega$.

K35 sets the interaction range.

The dynamics self-organise the mass spread to match it.

Not fully coupled. Not decoupled.

*At the threshold — where structure can propagate
but locality is preserved.*

This is not coincidence. This is self-consistency.

*δ chose the only scale at which space can exist:
large enough to connect, small enough to localise.*